

IBM Guardium Cryptography Manager

Protect sensitive data, mitigate risk, and prepare your organization for quantum resilience.



Highlights

Discovery and inventory – gain comprehensive overview

Policy and risk assessment – achieve faster compliance

Monitoring and reporting – turn insight into action

Certificate lifecycle management – prevent outages through automation

Remediation and access – accelerate quantum resilience with role-based control

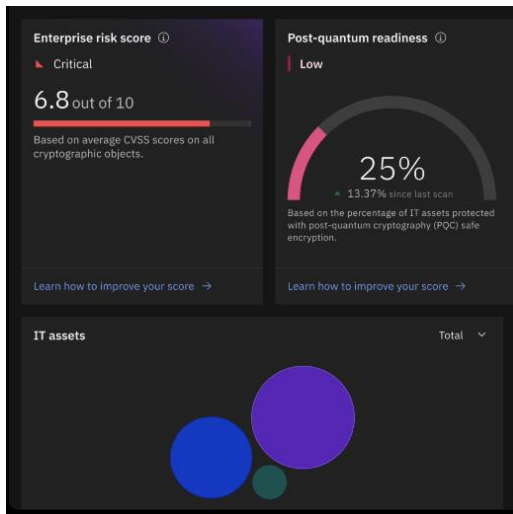
Organizations struggle to manage cryptography and protect sensitive data as quantum-safe deadlines approach and “harvest now, decrypt later” threats grow. Data sprawl across hybrid cloud and AI environments intensifies these risks, making it harder to locate and protect cryptographic assets. Beyond quantum threats, security teams face limited visibility, complex lifecycles, and fragmented tools. To keep pace with evolving regulations and emerging risks, organizations need cryptographic agility to rapidly discover, monitor, and update algorithms.

Crypto-agility – the ability to adapt cryptographic mechanisms in response to threats, technological advances, or vulnerabilities without disrupting infrastructure or business processes – is the solution to tackling today's challenges and building resilience against quantum threats.

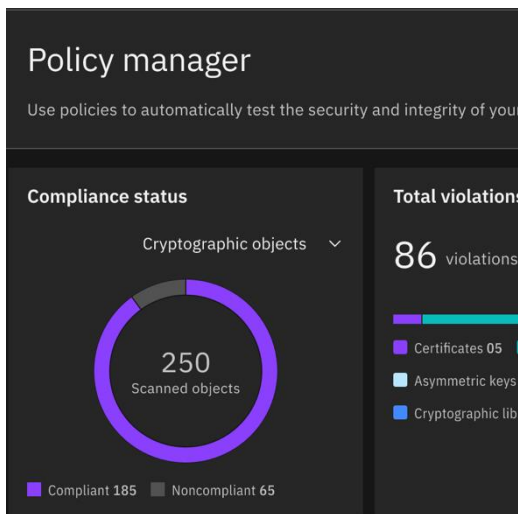
IBM Guardium® Cryptography Manager helps your organization achieve crypto-agility through discovery and inventory, risk assessment, cryptographic object lifecycle management, compliance readiness, and protection of cryptographic objects. It enables you to manage cryptography at scale, reduce risk, and move toward quantum-safe resilience—all from one solution.

With IBM Guardium Cryptography Manager you can:

- Discover and inventory your cryptographic landscape to gain complete visibility. Identify cryptographic objects, detect unknown or shadow cryptography, and map dependencies, ownership, and usage. Manage data sprawl by consolidating cryptographic assets across hybrid and AI-driven environments.
- Assess and comply using a policy engine aligned to compliance standards. Assess post-quantum cryptography risk, flag outdated or vulnerable algorithms and generate audit-ready reports.
- Manage lifecycle operations with automation that prevents outages. Centralize lifecycle operations in one tool, automate key generation and certificate renewal, and enforce consistent policies across the organization.
- Protect and adapt with a crypto-agile architecture that abstracts cryptography from applications, enabling organization-wide changes without disruption. Utilize encryption key management, assess the efficiency and feasibility of PQC algorithms, and protect against “harvest now, decrypt later” attacks.



Guardium Cryptographic Manager dashboard



Policies and compliance

Discovery and inventory – gain comprehensive view

IBM Guardium Cryptography Manager discovers data from diverse sources and formats to build a complete picture of your environment. It identifies IT assets and their associated cryptographic objects across the enterprise and presents key details such as type, subtype, location, and identifiers. The platform consolidates IT assets and cryptographic objects into a single inventory enriched with violation findings and cryptographic usage data from IBM Quantum Safe Explorer, network scans, and SSH public key discovery. It also supports import profiles for IBM Quantum Safe Explorer scan findings, enabling organizations to ingest and manage findings for code repositories and code assets, as well as custom transformers that ingest IT assets and cryptographic objects from custom file formats. In addition, Portfolio View provides visibility into repositories imported from Quantum Safe Explorer discovery findings, analytics findings, and CBOM data, helping teams identify vulnerable, non-quantum-safe, non-crypto-agile, and non-FIPS-compliant code repositories.

Plugin toolkit framework

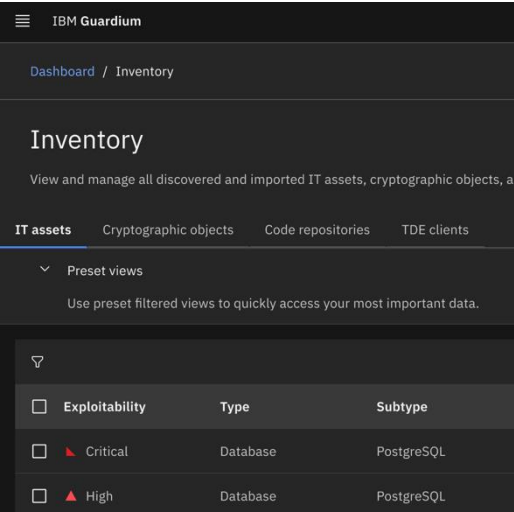
An open plug-in framework helps extend discovery and inventory by connecting to external systems and ingesting cryptographically relevant metadata into a standardized Guardium Cryptography Manager format. This helps organizations expand coverage faster across applications, infrastructure, PKI, certificates, keys, secrets management, SIEM, IAM, and HSM environments.

Policy and risk assessment – achieve faster compliance

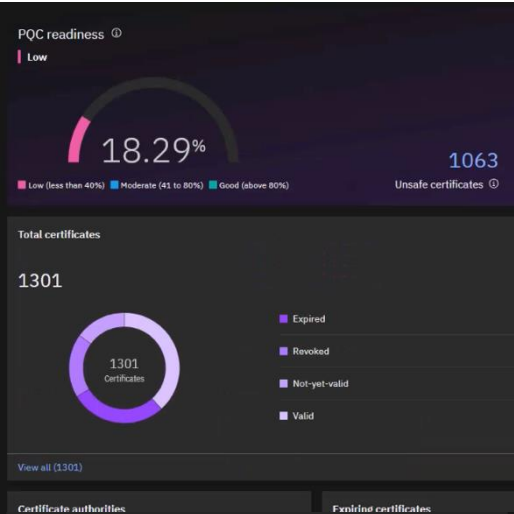
IBM Guardium Cryptography Manager creates and manages policies to evaluate risks and violations at scale. It includes system-defined policies aligned to National Institute of Standards and Technology guidance and supports custom rule-based policies through a built-in wizard. The solution analyzes assets and cryptographic objects for adherence to these policies, producing violations categorized as classical or post-quantum cryptography issues. Custom policies include compliance tags and control IDs, helping organizations map violations to internal or industry frameworks. Support for HIPAA, PCI-DSS, and GDPR expands compliance coverage and helps teams prioritize remediation. Filters, side-panel details, and exploitability explanations on violation pages provide added context for faster investigation and response.

Monitoring and reporting - turn insight into action

IBM Guardium Cryptography Manager continuously monitors cryptographic posture through interactive dashboards and exportable reports. Cryptographic posture, Certificate posture, and Compliance posture dashboards help teams track health, risk, violations, certificate status, and compliance coverage across IT assets and cryptographic objects. AI-driven insights support faster prioritization and remediation, while reports summarize inventory, violations, and overall risk for audits and executive briefings.



Inventory dashboard



Certificates dashboard

Certificate Lifecycle Management – prevent outages through automation

The solution provides unified visibility into certificate posture across the enterprise, reducing risk through centralized lifecycle oversight. It discovers and maps certificates, identifies weak or non-compliant ones, and pinpoints vulnerabilities such as expired or quantum-unsafe certificates. Using HashiCorp Vault’s PKI, IBM Guardium Cryptography Manager automates issuance, renewal, and revocation to minimize outages and manual errors. Built-in audit reporting and compliance automation support continuous alignment with regulatory standards. It also supports certificate auto-renewal at both the policy and Certificate Lifecycle Management level, deployment of certificates discovered through Kubernetes, and creation of CA-signed certificates for Venafi PKI to streamline renewal and remediation across more environments.

Transparent Database Encryption Key Management – solution for stronger data protection

The solution delivers seamless encryption key management across major databases, safeguarding critical business data from unauthorized access and breaches. Guardium Cryptography Manager leverages native encryption capabilities with centralized key lifecycle and policy management to simplify operations and strengthen compliance. By eliminating agent installation overhead, deployment becomes faster and more secure. Unified dashboards provide clear visibility into encryption key management posture, helping teams detect gaps, assess risk, and take timely remediation actions. With built-in automation and AI-driven insights, organizations can maintain continuous protection and compliance across their data landscape. It also supports Microsoft SQL Server Transparent Database Encryption (TDE) key management, extending centralized key management across more database environments.



Jira ticket creation

AI powered violation management

Remediation and access - accelerate quantum resilience with role-based control

IBM Guardium Cryptography Manager integrates with ticketing systems such as Jira and ServiceNow to streamline remediation workflows from detection through closure. Role-based access ensures appropriate control, with capabilities available to Super Administrator, Compliance Officer, Security Administrator, and Database Administrator roles. Fine-Grained Access Control (FGAC) provides more granular access management for IT asset groups and code repositories, helping ensure users can view only the resources they are authorized to access. Support for user groups, resource groups, and dynamic groups improves operational scale by allowing assets to be classified and updated automatically based on predefined rules and exceptions.

AI and automation – simplify remediation and operations

Guardium Cryptography Manager supports AI-driven and agent-based automation to help teams move faster from detection to action. It integrates with IBM watsonx.ai to generate AI-assisted summaries and ticket content for cryptographic violations, and it now supports agent integration through the out-of-the-box MCP Server. This allows teams to query GCM in natural language and automate tasks such as retrieving asset counts, identifying certificates nearing expiration, and listing active policy violations.

Flexible deployment and data ingestion

IBM Guardium Cryptography Manager supports installation through Helm on Kubernetes clusters or on Red Hat OpenShift Container Platform. It ingests data from a wide range of sources, including network scanners and Kubernetes; Vault for certificate discovery; Qualys and Nessus for IT and cryptographic assets; Cryptographic Bill of Materials (CBOM) for code assets; and IBM Quantum Safe™ Explorer for code vulnerabilities and post-quantum readiness assessments. It also supports preformatted file imports, with CSV uploads in the user interface and JSON ingestion through APIs for bulk data from external tools.

IBM Guardium Cryptography Manager system requirements

Storage class requirements

Nodes	Minimum of three worker nodes required for Monitors (MONs) and Object Storage Daemon (OSD)
Storage (per node)	At least one raw block device Hard Disk Drives (HDDs) or Solid state drives (SSDs) required for OSDs
Storage in GB	200
Storage capacity	100 GB to 1 TB per node (recommended starting point)
MONs	Three MONs (one per node)
Managers	Two Managers (one active and one standby)
OSDs	One to three OSDs per node (based on available disks)
Networking	Minimum of 1 Gb E, but 10 GbE is recommended for production. Bonding is also recommended.

Kubernetes and OCP cluster requirements based on sizing models

Sizing model	GCM services replicas	Middleware replicas	vCPU cores	Memory in GB	Storage in GB
Trial	1	1	24 cores	48 GB	150 GB
Small	2	3	8 cores for each of the three Control plane nodes	32 GB for each of the three Control plane nodes	Use 300 GB per Worker node
			16 cores for each of the three Worker nodes	64 GB for each of the three Worker nodes	
Large	3	3	16 cores for each of the three Control plane nodes	16 GB for each of the three Control plane nodes	Use 800 GB per Worker node
			32 cores for each of the three Worker nodes	64 GB for each of the three Worker nodes	

Browser recommendations:

Google Chrome incognito mode
Mozilla Firefox incognito mode

IBM Guardium Cryptography Manager is a unified, AI-powered solution that helps your organization protect sensitive data, mitigate risk, achieve crypto-agility, and build toward quantum resilience. It delivers comprehensive visibility through discovery and inventory and streamlines risk assessment and compliance. Prepare your enterprise for quantum resilient cryptography through crypto-agility with IBM Guardium Cryptography Manager.

For more information

To learn more about the solution, contact your IBM representative or IBM Business Partner.

© Copyright IBM Corporation 2026

Produced in the
United States of America
March 2026

IBM, the IBM logo, Guardium and Quantum Safe, are trademarks or registered trademarks of International Business Machines Corporation, in the United States and/or other countries. Other product and service names might be trademarks of IBM or other companies. A current list of IBM trademarks is available on [ibm.com/trademark](https://www.ibm.com/trademark).

This document is current as of the initial date of publication and may be changed by IBM at any time. Not all offerings are available in every country in which IBM operates.

THE INFORMATION IN THIS DOCUMENT IS PROVIDED "AS IS" WITHOUT ANY WARRANTY, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

IBM products are warranted according to the terms and conditions of the agreements under which they are provided.

